

De Morgan's Law

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

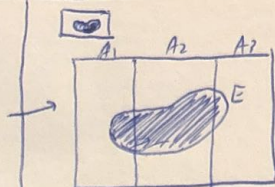
The Conditional Probability Formula

$$P(A \cap B) = P(A) P(B|A)$$

$$P(A \cap B \cap C) = P(A) P(B|A) P(C|A \cap B)$$

The Law of Total Probability

$$P(E) = \sum_{k=1}^n P(A_k) P(E|A_k)$$



Bernoulli Distribution

Models a single chance outcome

ex: one flip of one coin

$$X \sim \text{Bern}(p)$$

$$P(X=1) = p$$

success

$$P(X=0) = 1-p$$

fail

PMF:

$$P(X=x) = \begin{cases} p, & x=1 \\ 1-p, & x=0 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = p$$

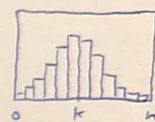
$$\text{Var}[X] = p(1-p)$$

Binomial Distribution

This is if you're doing a fixed # of Bernoulli experiments in a row. Ex: flipping 4 coins

What's the chance of winning exactly k times, $k \in [0, n]$?

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$



$$E[X] = np \quad \text{Var}[X] = np(1-p)$$

Bayes' Theorem

$$P(A_k|E) = \frac{P(A_k) P(E|A_k)}{\sum_{j=1}^n P(A_j) P(E|A_j)}$$

Negative Binomial Distribution

What if instead we want to keep running Bernoulli Experiments until we get the r th success?

$$X \sim \text{NB}(r, p)$$

$$P(X=k) = \binom{r-1}{k-1} p^r (1-p)^{k-r}$$

Probability Distribution aka

probability Mass Function PMF

$f(x_k) = P(X=x_k)$ gives the probability of each value

The collection $\{(x_k, P(X=x_k))\}_{k=1}^n$ is the probability distribution of X

Poisson Distribution

Moving on to continuous instead of discrete.

Suppose it's raining on a metal panel and on average, it gets λ drops a second.

What are the chances precisely k drops hit the panel?

$$P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad E[X] = \lambda \quad \text{Var}[X] = \lambda$$

$$E[X^2] = x_1^2 p_1 + \dots + x_n^2 p_n$$

Exponential Distribution

How long until the first drop hits? Call that time T

$$\text{CDF: } P(T \leq t) = 1 - e^{-\lambda t} \quad E[X] = \frac{1}{\lambda} \quad \text{Var}[X] = \frac{1}{\lambda^2}$$

$$\text{PDF: } \frac{d}{dt} \text{CDF} = \lambda e^{-\lambda t}$$

Gamma Distribution

How long until the k th drop?

$$P(T_k \leq t) = 1 - \sum_{k=0}^{r-1} \frac{(\lambda t)^k e^{-\lambda t}}{k!} \quad E[X] = \frac{r}{\lambda} \quad \text{Var}[X] = \frac{r}{\lambda^2}$$

A continuous RV X has a

PDF $f(x)$ if

$$f(x) \geq 0 \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

CDF then is

$$F(x) = \int_{-\infty}^x f(t) dt$$

Normal Distribution / Gauss Distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$E[X] = \mu \quad \text{Var}(X) = \sigma^2$$

Stirling's Approximation (large m)

$$m! \approx \sqrt{2\pi m} \left(\frac{m}{e}\right)^m$$

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

$$= E[X^2] - \mu^2$$

Uniform Distribution

$$\text{PDF } f(x) = \begin{cases} \frac{1}{b-a} & \text{on } [a, b] \\ 0 & \text{else} \end{cases}$$

$$\text{CDF } F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & \text{on } [a, b] \\ 1 & x > b \end{cases}$$

Beta Distribution?

α = observed successes
 β = " failures

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$\text{where } B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

Statistics - the science of making sense of data by collecting, analyzing, interpreting, and presenting it, turning raw data into meaningful info

Distribution - a function that shows all the possible values of a variable and how its spread. It provides info on:

- the range
- the central tendency (mean, median, mode)
- the variability (variance, std)
- the shape (symmetry, skewness)

Population vs Sample

- Hypotheses are made about the parameters of the population
- Hypotheses are tested using statistics from the sample

• parameters describe the population usually unknown
 mean μ , std σ

• statistics describe the sample
 $\Rightarrow$ used to estimate the true value
 sample mean $\bar{x}$, sample std s

• Frequency $\Rightarrow$ How often data pts appear in a dataset

Example frequency table	$x \backslash f(x)$
Dataset [2, 2, 3, 4, 4]	1 0 2 2 3 1 4 2 5 0

Probability is a measure of the likelihood of an event occurring.

$$P(A) = \frac{\# \text{ of } A\text{-success outcomes}}{\text{total } \# \text{ of outcomes}}$$

Probability Axioms

- 1) $\forall A_k, 0 \leq P(A_k) \leq 1$
- 2) $\sum_k P(A_k) = 1$
- 3) If A_k 's are disjoint,
 $P(\bigcup_k A_k) = \sum_k P(A_k)$

Types of Scales:

Nominal - no inherent order
 ex: U.S. state, gender, color

Ordinal - meaningful order, but intervals between them not uniform
 ex: Freshman, sophomore, ...

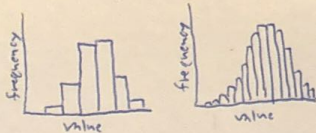
Interval - Numerical scale w/ equal intervals, but no true zero
 ex: SAT score
 F° or C° (interval)
 ratios don't have meaning
 "half as hot"

Ratio - Numerical w/ a true zero
 ex: Height, weight, distance

Discrete Variables $\rightarrow$ finite or countable
 ex: # cars

Continuous Variables $\rightarrow$ any value w/in a range
 ex: time 12.345

Histogram of continuous variable



The binning matters!

Outlier $\Rightarrow$ Significantly different from the rest of the population.

Either much lower or much higher often defined as falling outside of $1.5 \times$ the IQR or sometimes as more than $3 \times$ the std

Central Tendency

Median, Mean, Mode, Range

A Statistic takes a whole pile of data and reduces it to one number that represents what the data are like overall ... mean, median, std, etc

Mode $\rightarrow$ unimodal if just one
 $\rightarrow$ multimodal if a tie
 $\rightarrow$ no mode if all have same frequency

How to find the mean of a random variable X

• Finite Discrete Case

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

• Infinite Discrete Case

$$\mu = E[X] = \sum_{i=1}^{\infty} x_i p_i$$

where $p_i = P(X = x_i)$

• Continuous Case

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

$$\text{Range} = \text{Max} - \text{Min}$$

Interval $\rightarrow$ Mode valid
 $\rightarrow$ Median valid
 $\rightarrow$ Mean valid

Ordinal $\rightarrow$ Mode valid
 $\rightarrow$ Median valid

Nominal $\rightarrow$ Mode valid

Variability measures how much points in a dataset spread and differ from one another
 $\rightarrow$ range
 $\rightarrow$ quartiles
 $\rightarrow$ variance
 $\rightarrow$ std

Quantiles



$Q_1 \Rightarrow > 25\%$ of the data

$Q_3 \Rightarrow > 75\%$ of the data

Interquartile Range "IQR"

$$\Rightarrow Q_3 - Q_1$$

How to get Q_1 and such if that would give a fractional position?

Ex: $D = [1, 1, 2, 2, 2, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 6, 6, 6, 6, 6, 7, 7, 7, 8]$

$$\text{Len}(D) = 24$$

add 1

$$\text{Pos}(Q_1) = \frac{1}{4}(24+1) = 6.25$$

$$\text{Pos}(Q_2) = \frac{2}{4}(24+1) = 12.5$$

$$\text{Pos}(Q_3) = \frac{3}{4}(24+1) = 18.75$$

What to do w/ these?

$$\text{Value} = X_{LPJ} + (P - LPJ)(X_{HPJ} - X_{LPJ})$$

(note X_i is the first element)

$$Q_1 = X_6 + 0.25(3-3) = 3$$

$$\text{Variance} := s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

The $N-1$ is called Bessel's Correction.

When you compute variance from a sample, the data looks slightly less spread out than it really is. $N-1$ compensates for that

Standard Deviation, σ , is the

square root of variance.

It has the same units as mean

Set Theory

$IP := \{1, 2, 3, \dots\}$
 $IN := \{0, 1, 2, \dots\}$

For A^c to be defined, we need ϕ^c to be defined.

So all of it is within ϕ called the universal set

difference symbol

$A \setminus B :=$ stuff in A, but not in B

Inclusion-Exclusion Formula

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Power Set $P(S)$ is the set of all subsets of S

$$n(P(S)) = 2^{n(S)}$$

Binomial Coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, 0 \leq k \leq n$$

Pascal Relation

$$\binom{n-1}{k} + \binom{n}{k} = \binom{n}{k+1}$$

$$P(n, k) = \frac{n!}{(n-k)!}$$

You've got 8 books. How many ways to arrange 3 of them on a shelf?

$$P(8, 3) = \frac{8!}{(8-3)!}$$

You've got 8 books. How many ways to choose 3 of them to take on vacation?

$$\binom{8}{3} = \frac{8!}{3!(8-3)!}$$

How to make a 50/50 using an unfair coin?

HH (HT TH) TT
 restart

Expected Value

$$E = w_1 p_1 + \dots + w_n p_n$$

↓
 winnings

Final Exam Studying part 1

Main Topics:

- Variance and Covariance
- Normal Distribution and its properties
- Central Limit Theorem
- Point Estimation and Confidence Intervals
- Hypothesis Testing Procedure
- Type I vs Type II error

Variance

Variance is a measurement of how much a single variable spreads out around its mean.

$$\text{Var}(X) = E[(X - E[X])^2]$$

It's always ≥ 0

Large variance $\rightarrow$ wide spread

Small variance $\rightarrow$ tightly clustered.

A different version of the formula is

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Properties: $\text{Var}(aX + b) = a^2 \text{Var}(X)$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Variance

Example

Let's do an example using each of the two formulas to compute variance

X	1	2	3
P(X=x)	0.2	0.5	0.3

... Find the variance

~~Method 2:~~

~~Find $E[X]$. $E[X] = 1(0.2) + 2(0.5) + 3(0.3)$
 $= 2.1$~~

~~$\text{Var}(X) = E[(X - E[X])^2]$~~

part 1

X	1	2	3
P(X)	0.2	0.5	0.3

... find the variance

Method 1 - $\text{Var}(X) = E[(X - E[X])^2]$

• Find $E[X]$... $E[X] = 1(0.2) + 2(0.5) + 3(0.3) = 2.1$

• Compute each $x - E[X]$

x	$x - 2.1$	$(x - 2.1)^2$	P(x)	$(x - 2.1)^2 P(x)$
1	-1.1	1.21	0.2	0.242
2	-0.1	0.01	0.5	0.005
3	0.9	0.81	0.3	0.243
				total sum
				0.49

Method 2 - $E[X^2] - (E[X])^2$

$E[X^2] = 1^2(0.2) + 2^2(0.5) + 3^2(0.3) = 0.2 + 2 + 2.7 = 4.9$

$\Rightarrow E[X^2] - (E[X])^2 = 4.9 - 2.1^2 = 0.49$

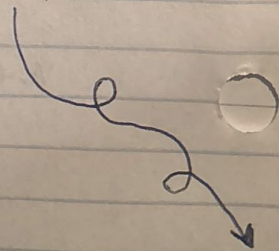
Continuous Variance

In the continuous case, the two formulas for variance become:

(1) $\text{Var}(X) = \int (x - \mu)^2 f(x) dx$

(2) $\text{Var}(X) = E[X^2] - (E[X])^2$, where $E[X] = \int x f(x) dx$
 $E[X^2] = \int x^2 f(x) dx$

Let's do an example



Final Exam Studying part 2

Suppose our distribution is $f(x) = 2x$, for $x \in [0, 1]$
(Note that it is valid since $\int_0^1 2x dx = 1$)

• Compute $E[X]$ $E[X] = \int_0^1 x \cdot 2x dx = \int_0^1 2x^2 dx = \left[\frac{2}{3} x^3 \right]_0^1 = \frac{2}{3}$

• Method 1:

$$\begin{aligned} \text{Var}(X) &= \int_0^1 (x - \mu)^2 f(x) dx = \int_0^1 \left(x - \frac{2}{3}\right)^2 2x dx = \int_0^1 \left(x^2 - \frac{4}{3}x + \frac{4}{9}\right) 2x dx \\ &= \int_0^1 2x^3 - \frac{8}{3}x^2 + \frac{8}{9}x dx = \left[\frac{1}{2}x^4 - \frac{8}{9}x^3 + \frac{4}{9}x^2 \right]_0^1 \\ &= \frac{1}{2} - \frac{8}{9} + \frac{4}{9} = \frac{9}{18} - \frac{16}{18} + \frac{8}{18} = \frac{1}{18} \end{aligned}$$

• Method 2:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = \int_0^1 x^2 \cdot 2x dx = \int_0^1 2x^3 dx = \left[\frac{1}{2} x^4 \right]_0^1 = \frac{1}{2}$$

$$\Rightarrow \text{Var}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

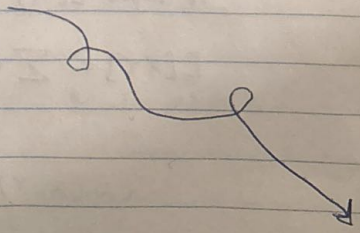
Covariance

Covariance measures how two variables move together.

It helps us assess whether or not the variables' deviations from the mean point in the same direction.

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Let's do an example

$$\left(\begin{array}{l} \text{or the shortcut} \\ \text{Cov}(X, Y) = E[XY] - E[X]E[Y] \end{array} \right)$$


part 2

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Suppose X & Y have the following joint distribution

	$X=1$	$X=2$	$X=3$
$Y=1$	0.3	0.2	0.1
$Y=2$	0.2	0.1	0.3

Covariance Visual Examples				
1	0.4	0	-0.4	-1

• Step 1: Compute $E[X]$ and $E[Y]$

$$E[X] = 1(0.3) + 2(0.3) + 3(0.4) = 0.3 + 0.6 + 1.2 = 2.1$$

$$E[Y] = 1(0.4) + 2(0.6) = 0.4 + 1.2 = 1.6$$

• Method 1: $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$

Compute the value for each (x, y) pair

Pair	prob	$x - E[X]$	$y - E[Y]$	$(x - E[X])(y - E[Y])$ prob
$x=1, y=1$	0.1	-1.1	-0.6	0.066
$x=2, y=1$	0.2	-0.1	-0.6	0.012
$x=3, y=1$	0.1	0.9	-0.6	-0.054
$x=1, y=2$	0.2	-1.1	0.4	-0.088
$x=2, y=2$	0.1	-0.1	0.4	-0.004
$x=3, y=2$	0.3	0.9	0.4	0.108
total sum				0.04

• Method 2: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$

To compute $E[XY]$, look at each (x, y) pair

$$E[XY] = \sum xy \cdot p(x, y) = \begin{aligned} &1 \cdot 1 \cdot 0.1 + 2 \cdot 1 \cdot 0.2 + 3 \cdot 1 \cdot 0.1 \\ &+ 1 \cdot 2 \cdot 0.2 + 2 \cdot 2 \cdot 0.1 + 3 \cdot 2 \cdot 0.3 \end{aligned} = 3.4$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 3.4 - 2.1 \cdot 1.6 = 0.04$$

Final Exam Studyly part 3

Note Covariance

Here are some other good covariance facts to know

- The relationship between variance and covariance is

$$\text{Var}(X) = \text{Cov}(X, X)$$

- Covariance depends on units

If you scale X or Y , covariance changes.

That's why we often use Correlation, which normalizes covariance:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

Continuous Covariance

If X and Y are both continuous, ~~we have~~ and we're given $f_{X,Y}(x,y)$,

~~$$E[X] = \int x f(x) dx$$~~

How to compute $E[X]$?

Option 1: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$

then $E[X] = \int x f_X(x) dx$

option 2: $E[X] = \iint x f_{X,Y}(x,y) dx dy$

Method 1:

$$\text{Cov}(X, Y) = \iint (x - E[X])(y - E[Y]) dx dy$$

Method 2:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y],$$

where $E[XY] = \iint xy f_{X,Y}(x,y) dx dy$

Half-Continuous Half-discrete Covariance

Suppose X is discrete & Y is continuous

$$\text{Cov}(X, Y) = \sum_i (x_i - \mu_X)(y - \mu_Y) f_{X,Y}(x_i, y) dy$$

or $\text{Cov}(X, Y) = E[XY] - E[X]E[Y],$

where $E[XY] = \sum_i \int x_i y f_{X,Y}(x_i, y) dy$

Variance and Covariance in Physics

Mean $\longleftrightarrow$ Center of Mass
 Variance $\longleftrightarrow$ Moment of Inertia
 Covariance $\longleftrightarrow$ Product of Inertia

Gaussian/Normal Distribution

If $X \sim N(\mu, \sigma^2)$, then the pdf is

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

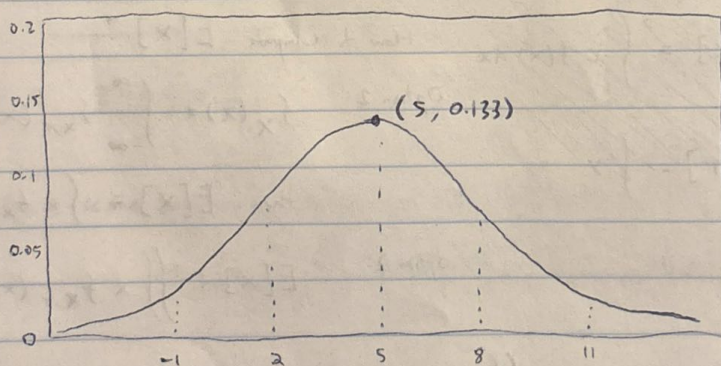
Most famously if $X \sim N(0, 1)$,

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Finding σ

From bell curve graph

If you're looking at the plot of a bell curve, how do you find σ from the image?



Option 1: Look for the inflection points (where it switches concavity)

These occur at $\mu \pm \sigma$.

In this case those are ≈ 2 & $8 \Rightarrow \sigma = 3$

Option 2: Use the max value

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \Rightarrow \overset{\text{max value}}{=} f(\mu) = \frac{1}{\sigma \sqrt{2\pi}}$$

$$\Rightarrow \sigma = \frac{1}{\overset{\text{max value}}{f(\mu)} \cdot \sqrt{2\pi}}$$

• Special case: For $X \sim N(0, 1)$,

$$\sigma = \frac{1}{\sqrt{2\pi}} \approx 0.399$$

Final Exam Studying part 4

Prove that
bell curve has
 $E[X] = \mu$

Prove that for distribution $X \sim N(\mu, \sigma^2)$ that $E[X] = \mu$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\left(\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Use u-substitution $z := \frac{x-\mu}{\sigma}$, $x = \mu + \sigma z$, $dx = \sigma dz$

$$E[X] = \int_{-\infty}^{\infty} (\mu + \sigma z) \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$= \mu \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{=1 \text{ because pdf}} + \sigma \underbrace{\int_{-\infty}^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{=0 \text{ because odd function}}$$

$$= \mu$$

Z-scores

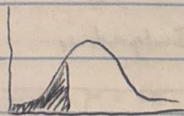
What is a z-score?

A z-score measures how many stds a value is away from the mean.

$$z := \frac{x - \mu}{\sigma} \quad \dots \text{It's a normalization technique}$$

What is a z-score table?

It gives the area ~~left~~ to the left of a given z-score



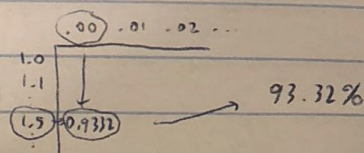
Example:

If the mean exam score is $\bar{x} = 70$ w/ std $s = 10$,

what's the probability of scoring below 85?

$$z = \frac{x - \bar{x}}{s} = \frac{85 - 70}{10} = 1.5$$

z-table



Linearity of Expectation

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$$E[aX + bY] = aE[X] + bE[Y]$$

What about for Variance?

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Independence

Random variables X & Y are said to be independent if

$$\forall x \in X, y \in Y, \quad P(X=x, Y=y) = P(X=x)P(Y=y)$$

X, Y, Z are independent if

$$\forall x, y, z, \quad P(X=x, Y=y, Z=z) = P(X=x)P(Y=y)P(Z=z)$$

Important properties:

If X & Y are independent, then

- $E[XY] = E[X]E[Y]$
- $\text{Cov}(X, Y) = 0$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- $g(X)$ & $h(Y)$ are also independent, $\forall g \in h$ functions

Note: Independence $\xRightarrow{\text{implies}}$ Cov = 0
Cov = 0 $\not\Rightarrow$ Independence

$\Rightarrow$ the classic example is
 $X \sim N(0, 1), Y = X^2$

Final Exam Study

part 5

Random Sampling

The random variables $X_1, X_2, \dots, X_n$ are said to form a random sample of size n if:

- (1) X_i are independent
- (2) All X_i come from the same distribution (aka identically distributed)

We call the X_i 's i.i.d.s independent and identically distributed

$\bar{X}$ is the sample mean $\rightsquigarrow \mu$

$\hat{p}$ is the sample proportion $\rightsquigarrow p$

s^2 is the sample variance $\rightsquigarrow \sigma^2$

Note: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

$\bar{X}$ is itself a random variable. We call its distribution

The sampling distribution of the sample mean

Sampling 2 Bell Curve

If $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$

then $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

(easy proof via $E\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$ and $\text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$)

Central Limit Theorem

The Central Limit Theorem

The above "almost" works even if X_i are not from a normal distribution!

So long as X_i are s.t. their mean & variance are finite, we have

as $n \rightarrow \infty$, $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow N(0,1)$ aka $\bar{X} \rightarrow N\left(\mu, \frac{\sigma^2}{n}\right)$

We can use the CLT reliably whenever $n \geq 30$ or so

CLT example

Example: A population has mean $\mu = 50$ & std $\sigma = 10$

The distribution is not normal / bell curve

You take a sample of size $n = 100$

What is the probability that sample mean is > 52 ?

$n = 100 \geq 30$, so we can use the CLT

$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(50, \frac{100}{100}\right)$

Normalize it $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{52 - 50}{1} = 2$

$P(\bar{X} > 52) = P(Z > 2) = \Phi(-2) \approx 2.28\%$

Point Estimates

What is a point estimate?

A point estimate is a single number you use to guess an unknown parameter

Examples include using the sample mean $\bar{X}$ to estimate the population mean μ

or s^2 to estimate σ^2

or $\hat{p}$ to estimate p

Often you'll see $\hat{\theta}$ to estimate an unspecified parameter θ

Statistic

~~Statistic~~ A statistic is any quantity computed from a sample of data.

$\bar{X}$, s^2 , $\hat{p}$ are all statistics

Unbiased Estimator

A statistic $\hat{\theta}$ is called an unbiased estimator of the parameter θ

if across all possible values of θ we have $E[\hat{\theta}] = \theta$

Examples: $\bar{X} = \frac{1}{n} \sum X_i$ is an unbiased estimator of μ
because $E[\bar{X}] = \mu$

Unbiased Estimator

Examples

- Let $X_1, X_2, \dots, X_n$ be a random sample from a population with mean $E[\bar{X}] = \mu$

Consider the following estimator for the parameter μ :

$$\hat{\theta} = \left(\frac{1}{n} \sum_{i=1}^n X_i \right) + a = \bar{X} + a$$

Is $\hat{\theta}$ a biased or unbiased estimator of the parameter μ ?

$$E[\hat{\theta}] = E[\bar{X} + a] \stackrel{\text{linearity}}{=} E[\bar{X}] + a$$

since $\bar{X}$ is an unbiased estimator of μ , $E[\bar{X}] = \mu$

For $a=0$, we have $E[\hat{\theta}] = \mu + a = \mu \implies$ unbiased

For $a \neq 0$, we have $E[\hat{\theta}] = \mu + a \neq \mu \implies$ biased

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Example: Let $X_1, \dots, X_n$ be a random sample from a population
with variance $\text{Var}(X_i) = \sigma^2$

Consider the following estimator for the parameter σ^2

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

Suppose that $E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2\right] = \sigma^2$

Is $\hat{\theta}$ a biased or unbiased estimator of the parameter σ^2 ?

$$E[\hat{\theta}] = E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2\right] \stackrel{\text{linearity}}{=} \frac{1}{n} \sum_{i=1}^n E[(X_i - \mu)^2]$$

$$\boxed{\text{recall that } E[(X_i - \mu)^2] = \text{Var}(X_i) = \sigma^2}$$

$$= \frac{1}{n} \sum_{i=1}^n \sigma^2 = \sigma^2 \Rightarrow \text{unbiased}$$

Example: Let $X_1, \dots, X_n$ be a random sample

w/ pdf $f(x) = \begin{cases} e^{-(x-\delta)} & x > \delta \\ 0 & \text{otherwise} \end{cases}$

Is the sample mean $\bar{X}$ a biased or unbiased estimator of δ ?

You may

Assume these facts: $\int_0^{\infty} x e^{-x} dx = 1$ $\frac{1}{\delta} \int_0^{\infty} e^{-x} dx = 1$

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

by formula for $E[X]$
for continuous

$$\int_{-\infty}^{\infty} x f(x) dx = \int_{\delta}^{\infty} x e^{-(x-\delta)} dx$$

change of variable

$$\int_0^{\infty} (y+\delta) e^{-y} dy \stackrel{\text{some calc later}}{=} 1 + \delta \neq \delta \Rightarrow \text{biased estimator}$$

Asymptotically
Unbiased
Estimator

Let $B_h(\theta) = E[\hat{\theta}] - \theta$ denote the bias of an estimator $\hat{\theta}$ based on a random sample of size n .

We say that $\hat{\theta}$ is an asymptotically unbiased estimator of θ

if and only if $\lim_{h \rightarrow \infty} B_h(\theta) = 0$

Sample Variance
Equation

Sample mean's equation is $\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$

Sample variance's equation is $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$

$$\begin{aligned} E[(x_i - \mu)^2] &= \sigma^2 \\ E[(\bar{X} - \mu)^2] &= \text{Var}(\bar{X}) \\ &= \frac{\sigma^2}{n} \end{aligned}$$

Good
Estimators

An estimator $\hat{\theta}$ of a parameter θ is desirable if it has the following properties:

- (i) Unbiased
- (ii) Asymptotic unbiasedness
- (iii) $\text{Var } \hat{\theta}$ should be as small as possible

Confidence
Intervals

A point estimator $\hat{\theta}$ gives a single number as an estimate of a parameter θ ,

but on their own they're a little useless.

What we need is $\hat{\theta} \pm \text{margin of error}$

If $\theta_1 \leq \theta \leq \theta_2$ are values of the random variables $\hat{\theta}_1 \leq \hat{\theta} \leq \hat{\theta}_2$ s.t.

for some specified probability $1 - \alpha$:

$$P(\theta_1 < \theta < \theta_2) = 1 - \alpha$$

Then we say the interval $\theta_1 < \theta < \theta_2$ is called a $((1 - \alpha) \cdot 100\%)$ Confidence Interval for θ .
 α is called the significance level

→ it's the probability of making a Type 1 error
(rejecting a true null hypothesis)

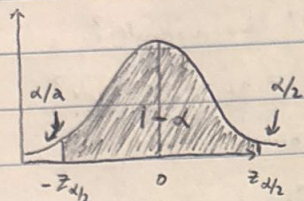
α represents the total probability outside the confidence interval

$1 - \alpha$ is called the degree of confidence

Final Exam Study

part 7

Confidence Intervals Theorem



If $\bar{X}$, the mean of a random sample of size n from a normal population w/ known variance σ^2 i.e. $N(\mu, \sigma^2)$, is used as an estimator of the population mean

Then the probability is $1-\alpha$ that

$$|\bar{X} - \mu| < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Z-table for the bell curve

Z-Table for the Standard Normal Distribution

α	$\alpha/2$	$z_{\alpha/2}$
0.10	0.05	1.645
0.05	0.025	1.960
0.02	0.01	2.326
0.01	0.005	2.575
0.002	0.001	3.090

Confidence Interval Example

Workers got a test. We don't know how they did on it

Sampled exactly $n=100$ of them

BUT! we do know that the std of their scores is $\sigma = 6.2$.

What can we assert w/ 99% confidence about the max error of our estimate?

$$1-\alpha = 0.99 \Rightarrow \alpha = 0.01 \Rightarrow \alpha/2 = 0.005 \xrightarrow[\text{chart}]{\text{use the above}} z_{\alpha/2} = 2.575$$

$$\text{The max error is } E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 2.575 \frac{6.2}{\sqrt{100}} \approx 1.30$$

Thus our $\pm$ is 1.30

Confidence Interval Theorem 2

If $\bar{X}$ is the value of the mean of a random sample of size n from a normal population w/ known variance σ^2 , then

$$\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

is a $(1-\alpha)100\%$ confidence interval

for the mean of the population.

Two-Sample Confidence Intervals

How do we compare two populations? Two machines? Two materials?

Call them populations 1 & 2. What if $n_1 \neq n_2$ in the sample?

What if $\sigma_1 \neq \sigma_2$? $\rightarrow$ assume they're normally distributed

We seek to create a confidence interval for $\mu_1 - \mu_2$

$$\text{Let } D = \bar{X}_1 - \bar{X}_2$$

$$\bar{X}_1 \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \quad \& \quad \bar{X}_2 \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right) \Rightarrow D \sim N(E[D], \text{Var}(D))$$

$$E[D] = \mu_1 - \mu_2$$

$$\text{Var}(D) = \text{Var}(\bar{X}_1 - \bar{X}_2) = (1)^2 \text{Var}(\bar{X}_1) + (-1)^2 \text{Var}(\bar{X}_2) + 2(1)(-1) \text{Cov}(\bar{X}_1, \bar{X}_2)$$

$\xrightarrow[\text{independent}]{\text{because}}$ $\text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2)$

$$\Rightarrow D \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Standardizing gives

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

$$P\left(|(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)| < z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right) = 1 - \alpha$$

Using CLT we can apply this formula for non-normal distributions too, so long as $n_1, n_2 \geq 30$

Two sample confidence interval Example

Suppose we got two kinds of lightbulbs

$N_1 = 36$ tested, $\bar{X}_1 = 410$ hours till burn out, $\sigma_1 = 24$ $N_2 = 49$, $\bar{X}_2 = 405$, $\sigma_2 = 21$

Construct a 95% confidence interval for $\mu_1 - \mu_2$

and decide if we can conclude if the 1st kind of bulb lasts longer.

$$\alpha = 0.05 \Rightarrow z_{\alpha/2} = 1.96 \text{ (using chart)}$$

The confidence interval for $\mu_1 - \mu_2$ is $\bar{X}_1 - \bar{X}_2 \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

Plug everything in and we get $-4.8 < \mu_1 - \mu_2 < 14.8$

This contains 0, so we cannot conclude
that the two lightbulbs have different
burn out times

Final Exam Studying part 8

Confidence Intervals for Proportions

Often when we take a sample study, we're not seeking to ~~point~~ estimate a mean but rather a proportion. How do those work?

We use The Binomial Model

Let $\theta = P(\text{success})$

$$X \sim \text{Bin}(n, \theta) \quad E[X] = n\theta \quad \text{Var}(X) = n\theta(1-\theta)$$

For large n , $X \approx N(n\theta, n\theta(1-\theta))$

$$\text{Standardization} \Rightarrow Z = \frac{X - n\theta}{\sqrt{n\theta(1-\theta)}} \quad (\text{recall } X \text{ is the \# of successes!})$$

If X is a binomial random variable w/ parameters n & θ

Suppose n is large and $\hat{\theta} = \frac{X}{n}$ (= experimental $\hat{p}$)

Then $\hat{\theta} \pm z_{\alpha/2} \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}$ is a $(1-\alpha) \cdot 100\%$ confidence interval for θ

Example

What % are defective? We test 200 of them and find 30 defective ones. Construct a 95% confidence interval for p aka θ

$$\hat{\theta} = \frac{30}{200} = 0.15, \quad \alpha = 0.05 \Rightarrow z_{\alpha/2} = 1.96$$

Using the formula, we get 0.15 ± 0.0494

The next couple lectures were all leading up to how to make a confidence interval for population variance from a sample.

To do that, we need deeper math tools

Moment

What is a moment?

A moment is just an expected value of a power of a random variable.

The first moment: $E[X] \Rightarrow$ This is the mean

The second moment: $E[X^2] \Rightarrow$ Related to spread, used in variance

The third moment: $E[X^3] \Rightarrow$ Related to asymmetry, skewness

Moment
Generating
Function

What is a Moment Generating Function?

It's a way to encode all the moments of a random variable into one function.

It's a compressed "fingerprint" of the distribution that contains the mean, variance, etc...

Start with the definition $M_X(t) = E[e^{tX}]$

The Taylor Expansion is $e^{tX} = 1 + tX + \frac{t^2 X^2}{2!} + \frac{t^3 X^3}{3!} + \dots$
 $M_X(t) = E[e^{tX}] = 1 + tE[X] + \frac{t^2}{2!} E[X^2] + \dots$

This crazy function generates the various moments!

First Moment / Mean $\Rightarrow M'_X(0) = E[X]$

Second Moment $\Rightarrow M''_X(0) = E[X^2]$

$$V_{X^2}(X) = M''_X(0) - (M'_X(0))^2$$

• Key property: $M_{X+Y}(t) = M_X(t) * M_Y(t)$

This all lets us convert sums into products

Sum of
squares



Chi-Squared
Distribution

Suppose we have n i.i.d.s $X_1, X_2, \dots, X_n \sim N(0, 1)$

If we set $X := X_1 + X_2 + \dots + X_n = \sum_{k=1}^n X_k$,

by the CLT we have $X \sim N(0, n)$

What if we instead have $X := X_1^2 + X_2^2 + \dots + X_n^2 = \sum_{k=1}^n X_k^2$

What kind of distribution does that have?

$\forall k \in [1, n]$, let $Y_k := X_k^2$

Find the Moment Generating Function of Y_k

$$M_{Y_k}(t) = E[e^{tY_k}] = E[e^{tX_k^2}]$$

Since $X_k \sim N(0, 1)$, its pdf is $\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

$$= \int_{-\infty}^{\infty} e^{tx^2} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \int_{-\infty}^{\infty} e^{-(\frac{1}{2} - t)x^2} dx$$

Final Exam Study

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continued... Use the known Gaussian Integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\frac{\pi}{\lambda}}$

$$\Rightarrow M_{Y_n}(t) = (1 - \lambda t)^{-1/2}$$

Now define $X = Y_1 + Y_2 + \dots + Y_n$ and recall $M_{A+B}(t) = M_A(t) + M_B(t)$

$$\Rightarrow M_X(t) = \prod_{i=1}^n (1 - \lambda t)^{-1/2} = (1 - \lambda t)^{-n/2}$$

We've learned squaring a normal gives a new distribution
and adding them changes the exponent.

We now have a whole family of distributions $(1 - \lambda t)^{-n/2}$

Let's give it a name

The distribution w/ MGF $(1 - \lambda t)^{-n/2}$ we call the Chi-Squared Distribution
w/ n degrees of freedom

and we denote it w/ the symbol $\chi^2(n)$

We've learned that $\sum_{k=1}^n X_k^2 \sim \chi^2(n)$

How this
connects to
sample variance

Suppose we have $X_i \sim N(\mu, \sigma^2)$

Standardize $Z_i = \frac{X_i - \mu}{\sigma} \sim N(0, 1)$

$$\sum_{i=1}^n Z_i^2 \sim \chi^2(n)$$

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \Rightarrow \sum_{i=1}^n (X_i - \mu)^2 \sim \sigma^2 \chi^2(n)$$

We don't know μ , so use $\bar{X}$,

but to do so we lose 1 degree of freedom

$$\sum_{i=1}^n (X_i - \bar{X})^2 \sim \sigma^2 \chi^2(n-1)$$

Sample variance $= s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{\sigma^2}{n-1} \chi^2(n-1)$

We know s^2
and can get
 χ^2 from
a table.
This lets us
solve for
 σ^2

Chi-Squared

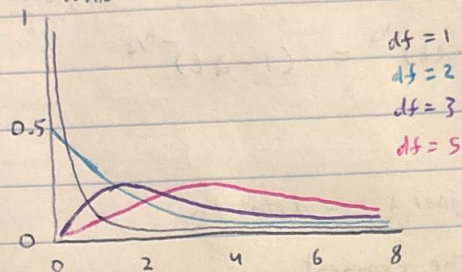
Facts

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\chi^2(n) \text{ has the complicated pdf } f(x) = \frac{1}{2^{n/2} \Gamma(n/2)} x^{n/2-1} e^{-x/2}$$

where $\Gamma(n) = (n-1)!$, for $n \in \mathbb{N}$

Which looks like:



$$\text{If } X = \sum_{i=1}^n X_i^2,$$

$$E[X] = n, \quad \text{Var}(X) = 2n$$

$$\text{If } X_1 \sim \chi^2(n) \text{ and } X_2 \sim \chi^2(m),$$

$$X_1 + X_2 \sim \chi^2(n+m)$$

Notation χ_n means $\chi^2(n)$

$\chi_{\alpha, n}$ means the chi-squared cutoff
w/ α significance level & n degrees of freedom

example $\chi_{0.05, 10}$ means look on a table

find 10 degrees of freedom.

Seek the value where 0.05 of the area is to the right

Using

Chi-Squared

for sample variance

$$\text{The important fact is } P\left(\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}\right) = 1-\alpha$$

↳ This is a $(1-\alpha) \cdot 100\%$ confidence interval
for σ^2

Sample

Variance

Example

We burn 12 rockets and find the sample variance to be 9 seconds².

Construct a 95% confidence interval for σ^2

We use $n=12$

$$s^2 = 9$$

$$\chi_{0.025, 11}^2 \quad \begin{array}{l} \text{from} \\ \text{table} \end{array} \quad 21.92$$

$$\chi_{0.975, 11}^2 \quad \begin{array}{l} \text{from} \\ \text{table} \end{array} \quad 3.816$$

Plug in:

$$\frac{(12-1)(9)}{21.92} < \sigma^2 < \frac{(12-1)(9)}{3.816}$$

$$\Rightarrow 4.52 < \sigma^2 < 25.94$$

Final Exam Studying

part 10

Hypothesis Testing

Steps in Hypothesis Testing

- (1) Formulate Null (H_0) and Alternative (H_a) Hypotheses
- (2) Choose a significance level (α), commonly $\alpha = 0.05$
- (3) Select an appropriate test statistic (ex: t-test)
- (4) Compute the p-value (using R) and compare to α
- (5) Make the decision
 - if $p > \alpha = 0.05$, Accept Null Hypothesis as true
 - if $p \leq \alpha = 0.05$, Reject the Null Hypothesis

$\alpha = 0.05 \rightarrow$ psychology
 $\alpha = 0.01 \rightarrow$ medical
 $\alpha = 10^{-7} \rightarrow$ physics

t-test

A one-sample t-test is used to determine whether the mean of a single sample is significantly different from a known or hypothesized population mean μ

Null Hypothesis: $\bar{X} = \mu$

Alt Hypothesis: $\bar{X} \neq \mu$ (two tailed)

or $\bar{X} > \mu$ (one tailed)

Definition of the t-value $t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$

The p-value is the probability of obtaining test results at least as extreme as the sample assuming the Null Hypothesis

Example

Claim: Mean is 1000. Sample: [1005, 1030, 1010, 1015, 1035, ..., 1015]

Null hypothesis: $\mu = 1000$

Alt hypothesis: $\mu \neq 1000$

Computed t-value $\Rightarrow t \approx 4.0435$

R-code output $t = 4.0435$, $df = 9$, $p\text{-value} = 0.002913$

95% confidence interval

1006.8, 1029.2

Does not contain 1000

$h-1$

$< 0.05 \Rightarrow$ reject the Null Hypothesis!

p-values

The p-value formula is $p\text{-value}(v, t) = \int_t^{\infty} \frac{\Gamma(\frac{v+1}{2})}{\sqrt{\pi v} \Gamma(\frac{v}{2})} \left(1 + \frac{x^2}{v}\right)^{-\frac{(v+1)}{2}} dx$

• Rule of thumb: Large t-values $\rightarrow$ small p-values

Student's t-table

You tend to find the p-value by looking at The Student's t-table

Ex: Find the p-value w/ $N=5$ and $t=1.5$

- $N=5 \Rightarrow \text{Degrees of Freedom} = N-1 = 4$
- Decide if you're use two-tailed (the default testing $\bar{x} = \mu$)
or one-tailed (testing $\bar{x} > \mu$)
- Look to the chart

2 tails α	0.5	0.4	0.2	0.1	0.05	0.02	0.01	0.002
4	0.74	0.94	1.53	2.1	2.8	3.7	4.6	7.2

Student t Distribution

When $X_1, \dots, X_n \sim N(\mu, \sigma^2)$,

we have $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

but what if we don't know σ , only an experimental S

we instead use $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1} \Rightarrow \text{Student's t-distribution w/ } n-1 \text{ degrees of freedom}$

Previously we had $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ Now we have $T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$

pdf of t_{n-1} distribution is $f_T(t) = \frac{\Gamma(\frac{v+1}{2})}{\sqrt{v\pi} \Gamma(\frac{v}{2})} \left(1 + \frac{t^2}{v}\right)^{-\frac{v+1}{2}}$, where $v := n-1$

How do we use t_{n-1} ?

- Hypothesis testing $H_0: \mu = \mu_0 \Rightarrow t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \Rightarrow \text{compare to table or we R}$
- Confidence Intervals $\bar{X} \pm t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}}$
- $t_{\alpha, n-1}$ means the cutoff value where α is in the right tail

Final Exam Studying

part II

Independent Sample Test

What do we do when we have two independent samples from two populations and we want to test if those populations have the same mean?

We use an independent sample test!

population 1 has mean μ_1 $\longrightarrow$ sample $\bar{X}_1, \bar{S}_1$

population 2 has mean μ_2 $\longrightarrow$ sample $\bar{X}_2, \bar{S}_2$

The independent-samples t-test compares the means of two independent groups to determine if they're significantly different

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)}}$$

Regular version assumes var. equal = True

Welch's assumes var. equal = False

They make different t-values because they have different #s for the degrees of freedom

Regular uses $df = n_1 + n_2 - 2$

Welch's uses a big gross approximation formula

We then use the t-table to get a p-value like before

or use R: ~~t.test~~

`two-samples_test_result <- t.test(A, B, alternative = "two.sided", var.equal = TRUE)`

Paired Samples t-test

A paired samples t-test is what you use when your two sets of data are not independent, but instead come in natural pairs.

Examples: Before and After a medicine

Twins

Same machine measured twice

The samples are linked like

$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$

We reduce the problem to one variable $D_i = X_i - Y_i$ and analyze like usual

It becomes a one-sample t-test

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

where $\bar{d} = \frac{1}{n} \sum (X_i - Y_i)$

R Code

~~library & power.t.test~~

Type I:

Type I Error (α): Rejecting a true null hypothesis (False Positive)

Type II errors

Type II Error (β): Failing to reject a false null hypothesis (False Negative)

What does α actually mean?

$\alpha = 0.05 \Leftrightarrow$ There is a 5% chance of falsely rejecting a true null hypothesis

It's a tradeoff: $\text{type I} \downarrow \Rightarrow \text{type II} \uparrow$ / $\text{type II} \downarrow \Rightarrow \text{type I} \uparrow$

Examples: Medical cancer test

Null Hypothesis H_0 : Patient is healthy

Type I (false positive) diagnose a healthy person as sick

Type II (false negative) miss the disease in a sick person

Court trial

Null Hypothesis H_0 : Defendant is innocent

Type I: convict an innocent person

Type II: let a guilty person off

Spam Filter

H_0 : Email is legitimate

Type I: Important email is marked as spam

Type II: Spam email gets through

Power

Power = $1 - \beta$... The probability of correctly rejecting H_0 when it is false

What increases power?

- larger sample size n
- larger effect size (difference in means)
- lower variability (smaller s)
- Larger α (easier to reject H_0)